

Mary Pletcher

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SYSTEMS AND INTERFACE ENGINEERING

Motivated and curious Systems Engineering Master’s graduate with a passion for tackling challenges and driving impact. Equipped with a foundation in UI/UX design, human factors engineering, programming fundamentals, and project management. Proven track record of independently solving evolving problems and collaborating effectively within teams to deliver exceptional value for users.

KEY COMPETENCIES

Design Thinking	Report Writing and Presenting	Usability Testing
UI/UX Design + Prototyping	Project Management	Survey Design and Analysis
Scripting and Automation	Requirements Gathering	3D + VR Prototyping

RELEVANT EXPERIENCE

Peraton - Systems Engineer	Summer 2024 - Present
- Led risk reduction data team by organizing and managing modeling and simulation data - Identified and tracked closure of approximately 30 risks to testing and development - Implemented digital optimizations and Sources of Truth using Powershell scripting, Jira, and Confluence	
Human IMPAC-T Lab - User Researcher and Lab Manager	Spring 2023 - Summer 2024
- Spearheaded 3 user experience research projects, balancing high level goals and interaction details - Taught an undergraduate course on applications of user research/design principles - Organized best practices and activities through Notion (digital project management tool)	
Virginia Tech Library - Prototyping Studio Consultant	Fall 2021 - Spring 2024
- Leveraged excellent customer service skills to support academic and personal projects - Developed educational materials for projects and tool training - Applied user-centered design for a children’s museum interactive installation project	
University of Nottingham, UK - Human Factors Researcher	Summer 2023
- Efficiently executed research study within short timeline, demonstrating organization and initiative - Programmed Unity app for VR/AR experience in simulated autonomous vehicle - Performing mixed methods data analysis on survey and think aloud protocol data using Python	
VR Immersive Training Project - Undergraduate Research Assistant	Spring 2022 - Spring 2023
- Collaborated with two students to create a literature review report on AR/VR ergonomic training - Prototyped VR + motion tracking training app based on application requirements	
Accenture Federal Services - Summer Analyst	Summer 2022
- Collaborated with solutions team on market/technology research presentations and graphics - Developed 4 decks, 1 one-pager, and supported work and key wins across 6 different proposals - Initiated and planned summer networking events with other summer interns	

EDUCATION & CERTIFICATIONS

M. Eng. in Industrial and Systems Engineering Virginia Tech (May 2024) Human Factors Engineering Track GPA: 3.91/4.00
Human-Computer Interaction Graduate Certificate Virginia Tech (May 2024)
B. S. in Industrial and Systems Engineering Virginia Tech (May 2023) Human Computer Interaction Minor GPA: 3.73/4.00

HIGHLIGHTED PROJECT

Extended Reality (XR) as a Product at Virginia Tech Graduate Lead (2023-2024)
Led a community effort to determine the business and educational value of XR products. Collected user requirements and advocated for user needs through surveys, focus groups, personas, market research, workflow maps, and system maps. Continuously reported findings to technology and teaching teams through update reports, documentation, and presentations.